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R V College of Engineering

R V Vidyanikethan Post
Mysuru Road Bengaluru - 560 059

IV Semester B.E Regular/Supplementary Examinations June/July-2025.
Common to EC/EI
Course : Signals and Systems-EC244AI

Time : 3 Hours

Maximum Marks : 100

Instructions to the students

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, and 9 and 10.

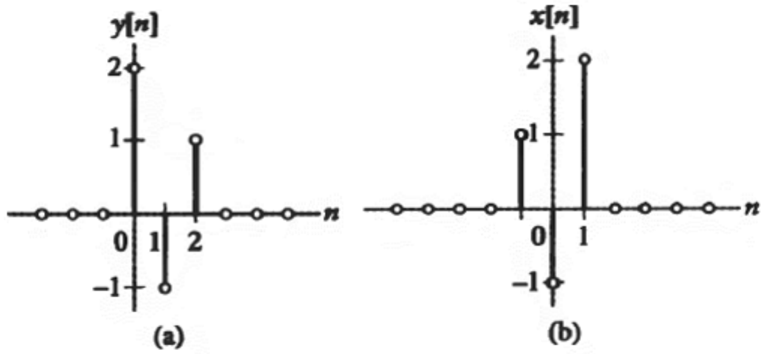
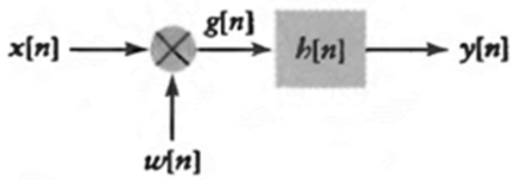
Part A

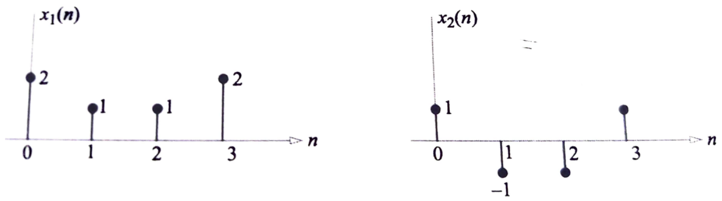
Question No	Question	M	CO	BT
1.1	Consider a discrete-time signal $x[n] = \sin(\pi^2 n)$, n being integer. Find whether the signal is periodic or not. If periodic, find its period.	02	2	4
1.2	A continuous-time signal $x(t) = 4\cos(200\pi t) + 8\cos(400\pi t)$, where t is in seconds, is input to a linear time-invariant (LTI) filter with the impulse response $h(t) = \begin{cases} \dfrac{2\sin(300\pi t)}{\pi t}, t \neq 0 \\ 600, t = 0 \end{cases}$ Let $y(t)$ be the output of this filter. Compute the maximum value of $\lvert y(t) \rvert$.	02	1	3
1.3	Determine the step response of the system if the overall impulse response $h[n]$ is the combination of $h_1[n]$ and $h_2[n]$ placed in cascade connection, where $h_1[n] = 2\delta[n] + 4\delta[n-2]$ and $h_2[n] = 3\delta[n+1] + 5\delta[n-1]$.	02	2	2
1.4	The two responses of an LTI system are given as $h_1(t) = e^{-2t}u(t)$ and $h_2(t) = e^{-t}u(t)$. Find the overall response $h(t)$ if the two systems are placed in cascade connection.	02	2	2

1.5	Evaluate the DTFS representation of the signal: $x(n) = \cos\left(\frac{6\pi n}{17} + \frac{\pi}{3}\right)$.	02	3	2
1.6	Let $x(n) = (1/2)^n u(n)$, $y(n) = x^2(n)$ and $Y(e^{j\Omega})$ to be the Fourier transform of $y(n)$ then find $Y(e^{j0})$.	02	3	2
1.7	DFT of a signals is $\cos^2 \Omega + \sin^2 2\Omega$. Determine the signal corresponding to the DFT.	02	3	2
1.8	In radix-2 FFT, if the number of points $N = 16$, then the number of complex additions and number of complex multiplications are respectively ____________ and ______________.	02	3	1
1.9	Discuss the impact of Zero Crossing Rate (ZCR) in signal processing techniques.	02	3	2
1.10	What is spectral leakage and how does it affect feature extraction?	02	4	1

Part B

Question No	Question	M	CO	BT
2a	Determine whether the following signals are energy signals, power signals, or neither. (i) $x(t) = e^{-at}u(t),\ a > 0$ (ii) $x(t) = tu(t)$ (iii) $x[n] = (-0.5)^n u[n]$ (iv) $x[n] = 2e^{j3n}$	08	1	2
2b	A discrete-time system is both linear and time-invariant. Suppose the output due to an input $x[n] = \delta[n]$ is given in the Figure (a) below. (i) Find the output due to an input $x[n] = \delta[n-1]$. (ii) Find the output due to an input $x[n] = 2\delta[n] - \delta[n-2]$. (iii) Find the output due the input depicted in the Figure (b).	08	1	3

				
3a	For the given impulse response $h[n] = u[n+1] – u[n-4]$, the LTI system is excited by the input signal $x[n] = u[n] – 2u[n-2] + u[n-4]$. Obtain the output response $y[n]$ and plot the same. Use graphical method to solve the problem.	10	2	4
3b	Find the step response of the system where the impulse response is given by $h(t) = tu(t)$. Plot the step response.	06	2	3
	OR			
4a	Compute the convolution integral: $u(t+1) * u(t-2)$. Plot the output response.	06	2	3
4b	Evaluate the following discrete time convolution sum analytically and plot $y[n]$, where, $y[n] = \beta^n u[n] * u[n-3]$, $\lvert\beta\rvert < 1$.	10	2	2
5a	Find the DTFS and DTFT representations of the following periodic signal. Sketch the magnitude and phase spectrum. $x[n] = 1 + \sum_{m=-\infty}^{\infty} \cos\left(\frac{\pi m}{2}\right)\delta[n-m]$	08	3	2
5b	The input to a discrete time LTI system is $x(n) = \cos\left(\frac{\pi n}{8}\right) + \sin\left(\frac{3\pi n}{4}\right)$. Use the DTFT to find the output of the system, $y(n)$, if the impulse response is $h(n) = \frac{\sin\left(\frac{\pi n}{4}\right)}{\pi n}$. Also sketch $X(e^{j\Omega})$, $H(e^{j\Omega})$ and $Y(e^{j\Omega})$.	08	3	2
	OR			
6a	Consider the system depicted in Figure. The input to a discrete time system is $x(n) = \delta(n) - \frac{\sin\left(\frac{\pi}{4}n\right)}{\pi n}$, $w(n) = (-1)^n$ and the impulse response $h(n) = \frac{\sin\left(\frac{\pi}{2}n\right)}{\pi n}$. 	10	4	4

	Use the DTFT to determine the output $y(n)$. Also sketch $G(e^{j\Omega})$, $X(e^{j\Omega})$, $H(e^{j\Omega})$ and $Y(e^{j\Omega})$.			
6b	Find the DTFS and DTFT representation for the following periodic signals, $x(n) = \cos\left[\frac{\pi}{8}n\right] + \sin\left[\frac{\pi}{5}n\right]$	06	3	2
7a	For the sequences, $x_1(n)$and $x_2(n)$ shown below:  (i) Compute the circular convolution of these sequences for $N = 4$ and $N = 8$. (ii) Compute the linear convolution of these sequences. (iii) What value of $N$is necessary so that linear and circular convolution are the same on the $N-$point interval? (iv) Determine the non-zero lengths of $x_1(n)$ and $x_2(n)$ and justify how could have found the results of part (iii) without performing the calculations.	08	3	4
7b	Assume that a complex multiplication takes 1 μs and that the amount of time take to compute $N-$point DFT is determined by the amount of time it takes to perform all of the multiplications. (i) How much time it takes to compute a 64-point DFT directly? (ii) How much time is required if an FFT is used? (iii) Repeat parts (i) and (ii) for a 256-point DFT.	08	3	1
	OR			
8a	Find the 4-point circular convolution of $x(n)$ and $h(n)$ as given below: $x(n) = \{1, 1, 1, 1\}$ and $h(n) = \{1, 0, 1, 0\}$ using radix-2 DIF-FFT algorithm.	10	3	3
8b	Two sequences $[a, b, c]$ and $[A, B, C]$ are related as,	06	4	4

	$\begin{bmatrix} A \\ B \\ C \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & W_3^{-1} & W_3^{-2} \\ 1 & W_3^{-2} & W_3^{-4} \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix}$ where, $W_3 = e^{j\frac{2\pi}{3}}$. If another sequence $[p, q, r]$ is derived as, $\begin{bmatrix} p \\ q \\ r \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & W_3^{1} & W_3^{2} \\ 1 & W_3^{2} & W_3^{4} \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & W_3^{2} & 0 \\ 0 & 0 & W_3^{4} \end{bmatrix} \begin{bmatrix} A/3 \\ B/3 \\ C/3 \end{bmatrix}$, then establish a relationship between the sequences $[p, q, r]$ and $[a, b, c]$.			
9a	A biomedical research team is developing an intelligent system to assist neurologists in diagnosing epilepsy. The system uses EEG (electroencephalogram) signals, which record brain activity through electrodes placed on the scalp. Since seizures are often reflected as sudden changes in signal characteristics, the team is working on extracting meaningful features from both time and frequency domains to train a classifier that can distinguish between normal and seizure episodes. (i) Explain how energy and skewness in the time domain could indicate seizure activity. (ii) Why might frequency domain features be better suited for detecting certain patterns in EEG data? (iii) How can combining both types of features improve the accuracy of seizure classification?	10	4	4
9b	Discuss how a spectrogram is derived using the Short-time Fourier transform.	06	2	3
OR				
10a	A tech startup is working on enhancing user experience in smart virtual assistants by recognizing the speaker's emotional state. They collect audio recordings of different users speaking under various emotional conditions such as happiness, sadness, anger, and surprise. By analyzing speech signals for patterns in both time and frequency, the team aims to extract features that reflect pitch, energy, and voice modulation, which can then be used to classify emotions in real time. (i) What role does the envelope of the signal play in identifying emotional states? (ii) Discuss how dominant frequency and peak values can be indicators of vocal pitch and intensity for emotion detection. (iii) Propose a basic pipeline involving feature extraction and classification for speech emotion recognition.	10	4	4

10b	Discuss the importance of mean. variance, and standard deviation in time-domain feature extraction. How do these features help in distinguising between different types of singals? Provide an example.	06	2	3